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Prognosis of TiO_2 abundance in lunar soil using a non-linear analysis of Clementine and LSCC data

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Abstract

We suggest a technique to determine the chemical and mineral composition of the lunar surface using artificial neural networks (ANNs). We demonstrate this powerful non-linear approach for prognosis of TiO_2 abundance using Clementine UV–VIS mosaics and Lunar Soil Characterization Consortium data. The ANN technique allows one to study correlations between spectral characteristics of lunar soils and composition parameters without any restrictions on the character of these correlations. The advantage of this method in comparison with the traditional linear regression method and the Lucey et al. approaches is shown. The results obtained could be useful for the strategy of analyzing lunar data that will be acquired in incoming lunar missions especially in case of the Chandrayaan-1 and Lunar Reconnaissance Orbiter missions.
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Keywords: The moon; Lunar surface; Chemical composition; Titanium abundance; Prognosis; Artificial neural network

1. Introduction

Investigation of the Moon requires a determination of its surface chemistry and mineralogy with remote-sensing methods. A number of remote-sensing techniques based on γ-ray and neutron flux measurements, X-ray and optical spectroscopy are used to acquire the chemistry and mineralogy of the uppermost layers of the lunar regolith. At present time the optical spectroscopy onboard spacecrafts orbiting the Moon suggests both high spatial resolution and global coverage for multispectral albedo. The spectral properties of lunar soils are directly linked to its chemistry and mineralogy (e.g., Adams and McCord, 1971, 1972). The interpretation of spectral properties of the lunar surface relies on principles derived directly from mineral physics (e.g., Burns, 1993). Massive spectrophotometric studies with high spectral resolution in the UV–VIS–NIR range with use of ground-base telescopes (e.g., McCord et al., 1981; Pieters, 1986, 1993; Hawke et al., 2003) proved a possibility to determine fundamental characteristics of major diagnostic absorptions in lunar minerals. Establishing the direct correlation of blue-visible color ratio with TiO_2 content (Charette et al., 1974; Pieters, 1978; Jaumann, 1991; Johnson et al., 1991) was a classic example of applying the spectrophotometry to identify the composition of the lunar surface. The compositional information from samples delivered from nine isolated locations (Apollo-11, 12, 14–17, and Luna-16, 20, 24 stations) was extrapolated to the entire globe attempting evaluation of the Moon's geochemical diversity. Lunar samples are representative of Apollo and Luna landing sites, and do not include examples of unsampled soils that cover large areas of the Moon such as the young high-Ti basalts of the western portion of the lunar nearside (e.g., Pieters, 1978). These basalts are believed to be olivine-rich (Staid and Pieters, 2001). The lack of samples decreases the reliability of global prognoses based on the chemical–optical relationships for available lunar samples.

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The global compositional evaluation indicates broad regional variations of the abundance of TiO_2 in lunar basalts. The abundance varies from <1 wt% to >14 wt% TiO_2 (Charette et al., 1974; Neal and Taylor, 1992; Blewett et al., 1997; Lucey et al., 2000), thus the concentration of TiO_2 is often used for classifying the lunar mare basalts. Overall the evaluation of the TiO_2 content in lunar soils and rocks is a key problem for understanding the origin and evolution of the mantle and crust materials. The Clementine spectral reflectance (CSR) data (Nozette, 1994) are widely used for chemical and mineralogical interpretation due to their global coverage and 100 m/pix spatial resolution. In the past decade after Clementine several new TiO_2 estimation approaches have been presented.

The titanium maps derived from optical spectroscopy have been improved due to comparison with those obtained by the analysis of Lunar Prospector (LP) γ-ray and neutron flux distributions (Elphic et al., 2002). For example, LP data were used as ground-truth to calibrate CSR optical data in terms of titanium content (Shkuratov et al., 2005b). However, the spatial resolution of LP chemical maps is much lower than that for optical data, thus it prevents the direct comparison of two datasets for individual sampling sites and stations on the Moon.

Using samples data Lucey et al. (1998, 2000) evaluate the TiO_2 content with an angular parameter on the correlation diagram "color ratio UV/VIS versus VIS reflectance". For calibration they used data for Apollo landing sites and sampling stations. The so-called Ti-sensitive optical parameter

$$\Theta_{\mathrm{Ti}} = \arctan\{[(A_{415}/A_{750}) - y]/[A_{750} - x]\}, \qquad (1)$$

where A_{415} and A_{750} are reflectances at 415 and 750 nm, $x=0.00$ and $y=0.42$, was proposed as a measure of TiO_2 content (Lucey et al., 2000). Here x, y is an empirically defined position of dark and spectrally neutral ilmenite end member in the UV/VIS—VIS diagram for samples. The power-law function was proposed as the best fit for TiO_2 abundance

$$\mathrm{TiO_2, wt\%} = A\Theta_{\mathrm{Ti}}^{B}, \qquad (2)$$

where empirical parameters $A = 3.708$, $B = 5.979$ were optimized for the best fit of the relationship "TiO_2, wt% versus Θ_{Ti}" (Jolliff, 1999). Eqs. (1) and (2) have been obtained with comparing the TiO_2 abundance for returned lunar soils and Clementine spectral measurements of the Apollo sampling stations. We note that only 2 from 5 spectral bands available in CSR UV–VIS dataset are used for calculation of TiO_2 abundance. As a possible consequence of using 2 bands only Lucey's approach predicts too high TiO_2 values for maria (up to 18%) unlike mare samples available. The comparison with other techniques used $\geqslant 3$ Clementine bands (Shkuratov et al., 2003a) also shows that the scale of TiO_2 distribution obtained by Lucey et al. (2000) is noticeably stretched. As one can see in Fig. 1, Lucey's regression describes well these data, if Apollo-11 and Luna-16, 24 samples are excluded from the analysis. The TiO_2 map based on Lucey's approach is presented in Fig. 2 and clearly reveals TiO_2 values for maria as high as 18 wt% (cf. Fig. 1 for returned mare samples). We also note that the TiO_2 distribution derived with Eq. (2) for mare soils appears unimodal (Fig. 3) unlike the distribution for "ground-truth" returned samples (see below).

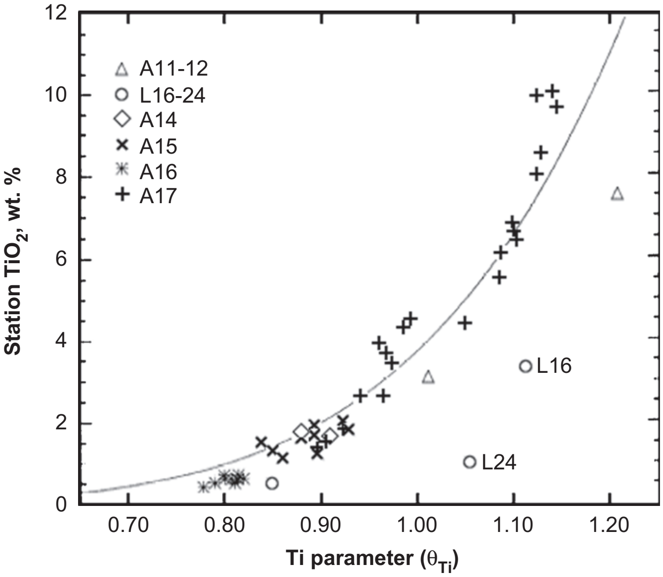

Fig. 1. TiO_2 abundance of lunar soils versus the spectral titanium parameter Θ_{Ti} derived from Clementine measurements: Lucey's approach (Eq. (2)) (adopted from Lucey et al. (2000)).

Gillis et al. (2003) have revised Lucey's method. They performed an analysis of data for the three sites (Apollo-11 and Luna-16, 24) deviating significantly from the others. They suggested subdividing the lunar surface into two classes related to Apollo-11, Luna-16, 24 landing sites and the rest ones, and applied two approximation functions (see Fig. 4). As result the TiO_2 concentration distribution became bimodal. Unfortunately, this assumption leads to the problem of separation of the lunar surface into two classes: one can expect the discontinuity of TiO_2 maps at the border of two optical classes. Moreover, it looks very doubtful to use only 4 points to obtain the approximation with three free parameters for the second regression (dotted curve in Fig. 4).

The two discussed methods for TiO_2 prediction operate only with two reflectances (415, 753 nm), neglecting the other three (899, 952, and 1000 nm), which also may be affected by titanium. We also note that these techniques use different sets of data to establish the relations in Figs. 1 and 4: chemical data for selected lunar samples (1 mm–1 cm in size) and orbital optical data (the element of resolution is 100 m). This assumes high homogeneity of the sampling areas and high representativeness of samples. An important factor affected the reliability of TiO_2 prognosis using spectrophotometry is the impossibility to reproduce in laboratory the pristine regolith microstructure which is

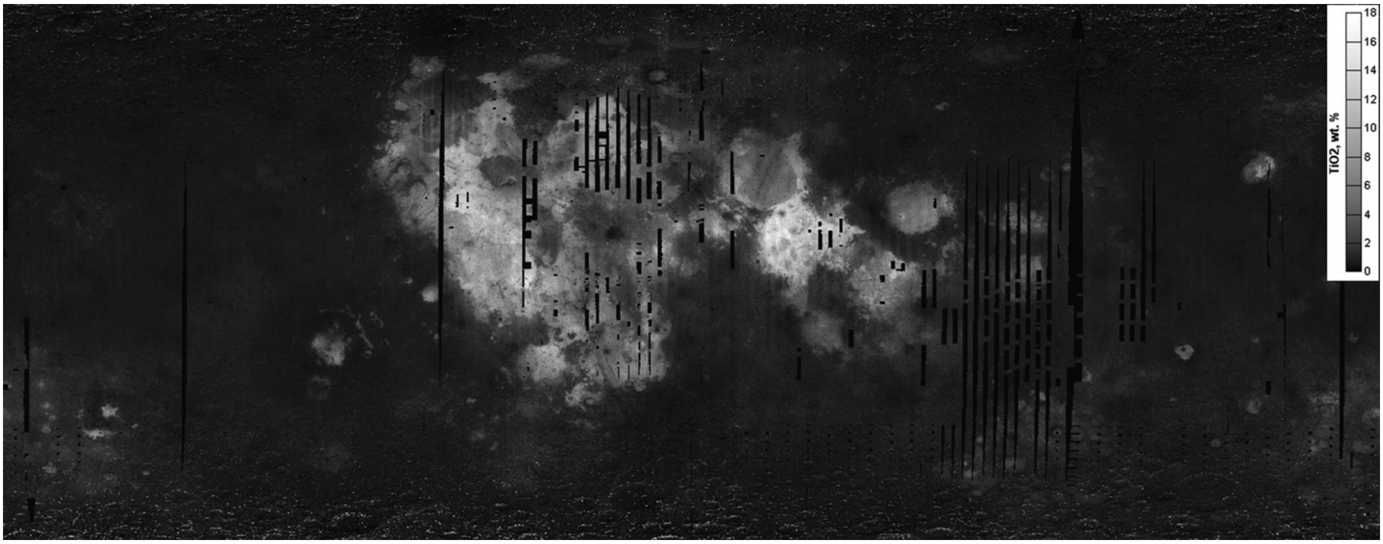

Fig. 2. Map of TiO_2 abundance corresponding to Lucey's prognosis (Eq. (2)). Projection is simple cylindrical; the latitude range is ±70°.

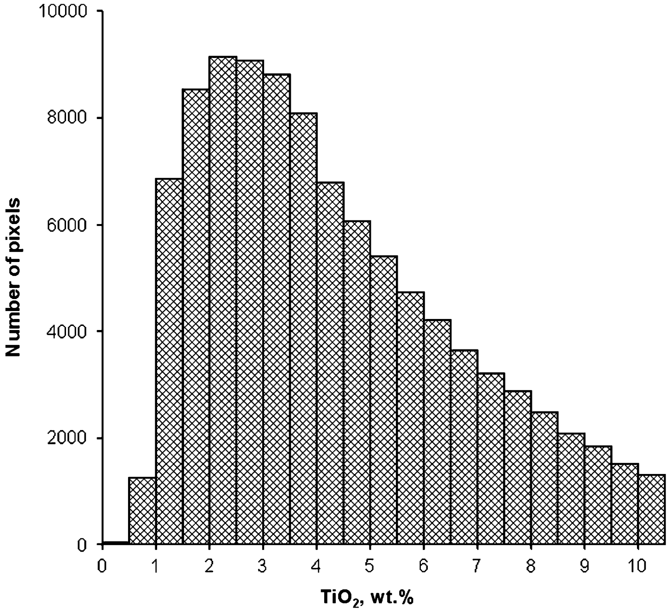

Fig. 3. Histogram of TiO_2 concentrations built for mare regions (reflectance <0.14 at 750 nm band) using Lucey's prognosis. The TiO_2 map with 8 km/pix resolution is used.

formed in vacuum under cosmogenic factors; this structure is destroyed when the samples are delivered to the Earth. This sub-millimeter structure may introduce differences to the shadow hiding effect in the photometric function (Shkuratov et al., 2001).

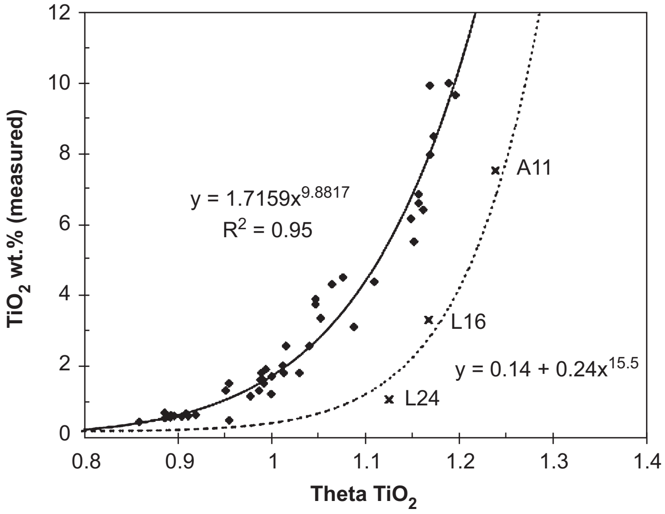

Fig. 4. TiO_2 abundance of lunar soils versus the spectral titanium parameter Θ_{Ti} derived from Clementine measurements (Gillis et al., 2003).

The different way to make such a prognosis is to use a simultaneous analysis of spectral and chemical data for lunar samples in order to build a calibration and then to use this for remote-sensing data. The Lunar Soil Characterization Consortium (LSCC) data suggest a unique opportunity to study direct links between spectral characteristics and the chemical/mineral composition of lunar soils (Pieters et al., 2000, 2002, 2006; Taylor et al., 2001). These links can be used for a prognosis of the lunar surface composition with spectrophotometric observations. A key point is that Clementine data and LSCC spectral measurements of lunar soils use the same photometric calibration (Pieters, 1999). The method based on the use of LSCC spectro-chemical sample data (Pieters et al., 2002; Shkuratov et al., 2003a, b) exploits a multiple linear regression (MLR). It is assumed that TiO_2 abundance linearly correlates with the spectral parameters:

$$\log(\mathrm{TiO_2}) = w_0 + \sum_{i=1}^{6} w_i a_i, \tag{3}$$

where w_0 and w_i are the weight coefficients, a_i are the spectral parameters. The following set of 6 spectral parameters

Table 1
Spectral and composition data for lunar soils used in this paper

Sample	Size (μm)	*A*(415)	*A*(750)	*A*(900)	*A*(950)	*A*(1000)	TiO_2%	I_s/FeO
10084	<10	0.0462	0.0793	0.0900	0.0928	0.0958	7.25	145.0
10084	10–20	0.0431	0.0691	0.0746	0.0751	0.0767	7.94	87.0
10084	20–45	0.0437	0.0645	0.0638	0.0626	0.0627	8.30	67.0
10084	Bulk	0.0448	0.0726	0.0790	0.0794	0.0813	7.54	88.0
12001	<10	0.0566	0.1093	0.1235	0.1274	0.1328	2.78	115.0
12001	10–20	0.0456	0.0822	0.0848	0.0847	0.0882	2.96	67.0
12001	20–45	0.0465	0.0759	0.0719	0.0702	0.0725	3.20	51.0
12001	Bulk	0.0496	0.0868	0.0912	0.0919	0.0954	2.96	62.0
12030	<10	0.1357	0.2066	0.2165	0.2176	0.2252	3.01	32.0
12030	10–20	0.0994	0.1541	0.1350	0.1295	0.1359	3.32	17.0
12030	20–45	0.0995	0.1474	0.1211	0.1148	0.1193	3.74	12.0
12030	Bulk	0.1101	0.1683	0.1572	0.1535	0.1607	3.23	20.0
15041	<10	0.0534	0.1121	0.1274	0.1317	0.1377	1.79	161.0
15041	10–20	0.0418	0.0762	0.0792	0.0793	0.0830	1.88	92.0
15041	20–45	0.0460	0.0760	0.0728	0.0729	0.0764	2.03	66.0
15041	Bulk	0.0461	0.0867	0.0917	0.0934	0.0973	1.83	93.0
15071	<10	0.0558	0.1137	0.1285	0.1321	0.1375	1.57	159.0
15071	10–20	0.0472	0.0876	0.0880	0.0876	0.0924	1.88	80.0
15071	20–45	0.0496	0.0856	0.0798	0.0799	0.0835	2.33	49.0
15071	Bulk	0.0501	0.0961	0.1003	0.1024	0.1068	1.81	71.0
70181	<10	0.0554	0.0989	0.1156	0.1207	0.1257	6.54	104.0
70181	10–20	0.0513	0.0803	0.0879	0.0893	0.0923	7.88	63.0
70181	20–45	0.0472	0.0705	0.0722	0.0724	0.0743	8.11	53.0
70181	Bulk	0.0488	0.0783	0.0862	0.0878	0.0908	7.57	61.0
71061	<10	0.0933	0.1551	0.1942	0.2031	0.2096	7.89	28.0
71061	10–20	0.0753	0.1162	0.1256	0.1278	0.1316	8.94	14.0
71061	20–45	0.0769	0.1151	0.1007	0.0975	0.0983	9.48	9.0
71061	Bulk	0.0882	0.1325	0.1524	0.1569	0.1610	8.76	16.0
71501	<10	0.0464	0.0842	0.0986	0.1018	0.1050	8.27	88.0
71501	10–20	0.0463	0.0753	0.0812	0.0810	0.0831	9.83	50.0
71501	20–45	0.0469	0.0749	0.0738	0.0716	0.0727	10.70	28.0
71501	Bulk	0.0465	0.0767	0.0838	0.0846	0.0867	9.31	44.0
79221	<10	0.0520	0.0897	0.1041	0.1082	0.1131	5.83	169.0
79221	10–20	0.0457	0.0711	0.0776	0.0794	0.0823	7.21	78.0
79221	20–45	0.0469	0.0687	0.0690	0.0697	0.0714	7.38	57.0
79221	Bulk	0.0491	0.0791	0.0887	0.0916	0.0949	6.39	91.0
14141	<10	0.2182	0.3280	0.3352	0.3366	0.3475	1.51	14.5
14141	10–20	0.1702	0.2590	0.2393	0.2385	0.2559	1.71	11.6
14141	20–45	0.1370	0.2009	0.1598	0.1565	0.1726	1.96	5.8
14141	Bulk	0.1780	0.2623	0.2444	0.2441	0.2596	1.65	9.7
14163	<10	0.1134	0.1970	0.2175	0.2227	0.2313	2.07	87.0
14163	10–20	0.0804	0.1391	0.1442	0.1459	0.1542	1.88	64.8
14163	20–45	0.0613	0.1022	0.0937	0.0946	0.1000	2.00	43.2
14163	Bulk	0.0903	0.1540	0.1632	0.1675	0.1741	1.90	66.5
14259	<10	0.0769	0.1564	0.1789	0.1862	0.1937	2.02	174.8
14259	10–20	0.0594	0.1184	0.1301	0.1340	0.1420	1.96	101.8
14259	20–45	0.0492	0.0809	0.0819	0.0826	0.0870	1.99	77.2
14259	Bulk	0.0604	0.1146	0.1253	0.1291	0.1359	1.80	108.6
14260	<10	0.0659	0.1375	0.1588	0.1649	0.1715	1.94	144.9
14260	10–20	0.0575	0.1179	0.1293	0.1330	0.1400	1.98	98.9
14260	20–45	0.0465	0.0793	0.0788	0.0799	0.0849	1.86	80.2
14260	Bulk	0.0574	0.1109	0.1210	0.1241	0.1321	1.85	93.3
61141	<10	0.1361	0.2464	0.2746	0.2827	0.2921	0.59	119.3
61141	10–20	0.1032	0.1818	0.1963	0.2012	0.2090	0.64	81.6
61141	20–45	0.0763	0.1265	0.1350	0.1388	0.1448	0.58	75.5

Table 1 (*continued*)

Sample	Size (μm)	A(415)	A(750)	A(900)	A(950)	A(1000)	TiO_2%	I_s/FeO
61141	Bulk	0.1138	0.1960	0.2131	0.2190	0.2271	0.59	94.5
61221	<10	0.3171	0.4216	0.4351	0.4377	0.4440	0.50	19.8
61221	10–20	0.2976	0.3887	0.3844	0.3860	0.3939	0.54	13.9
61221	20–45	0.2455	0.3181	0.2846	0.2814	0.2896	0.56	8.4
61221	Bulk	0.3020	0.3866	0.3821	0.3825	0.3906	0.52	13.6
62231	<10	0.1333	0.2350	0.2641	0.2721	0.2810	0.58	169.0
62231	10–20	0.1156	0.1952	0.2114	0.2164	0.2238	0.61	109.9
62231	20–45	0.1043	0.1492	0.1590	0.1628	0.1680	0.58	80.7
62231	Bulk	0.1110	0.1858	0.2037	0.2096	0.2169	0.60	116.7
64801	<10	0.1403	0.2551	0.2854	0.2939	0.3038	0.61	115.2
64801	10–20	0.1124	0.2014	0.2202	0.2265	0.2350	0.68	84.9
64801	20–45	0.1023	0.1752	0.1865	0.1911	0.1986	0.63	83.4
64801	Bulk	0.1139	0.1965	0.2143	0.2205	0.2288	0.65	92.2
67461	<10	0.2780	0.4133	0.4307	0.4350	0.4475	0.34	35.2
67461	10–20	0.2275	0.3382	0.3347	0.3375	0.3518	0.39	23.9
67461	20–45	0.1934	0.2801	0.2773	0.2796	0.2922	0.44	22.3
67461	Bulk	0.2493	0.3596	0.3607	0.3651	0.3795	0.35	29.8
67481	<10	0.2550	0.3899	0.4138	0.4200	0.4324	0.42	38.5
67481	10–20	0.2116	0.3192	0.3261	0.3309	0.3444	0.40	33.0
67481	20–45	0.1633	0.2336	0.2368	0.2392	0.2490	0.49	20.7
67481	Bulk	0.2264	0.3243	0.3343	0.3403	0.3525	0.44	33.5

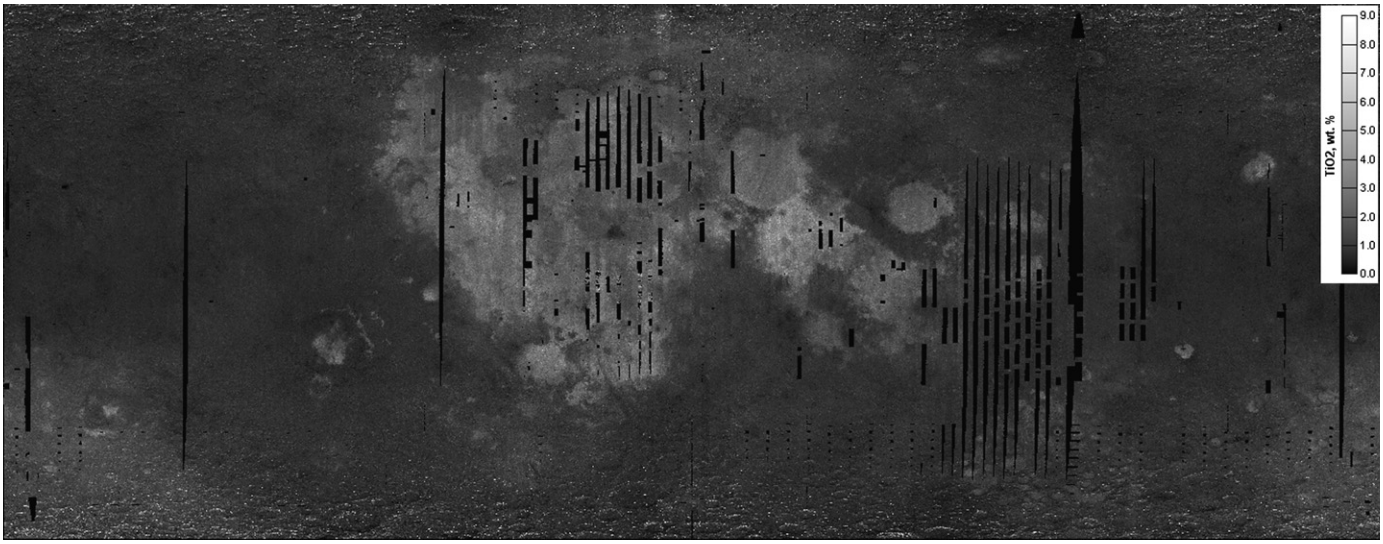

Fig. 5. Map of TiO_2 abundance prognosis using MLR (Eq. (3)).

was used in Shkuratov et al. (2003a, b): reflectance $A_R = A(750\,\text{nm})$, color-indices: $C_{BR} = A(415)/A(750)$, $C_{IR1} = A(900)/A(750)$, $C_{IR2} = A(950)/A(750)$, $C_{IR3} = A(1000)/A(750)$, and bend $D = A(750)A(1000)/[A(900)]^2$ (this set provides a higher correlation with composition parameters in comparison with initial CSR UV–VIS data). Logarithm in Eq. (3) is used to avoid negative values of the predicted parameter. The weight coefficients are found using the minimal least-squares method for the LSCC data (Table 1). This method was applied to CSR UV–VIS data. Such a technique gives good results for some composition components, e.g., pyroxene content and maturity degree (Shkuratov et al., 2005a, 2007; Pieters et al., 2006). This is not the case for TiO_2. The result of MLR prognosis of TiO_2 abundance is shown in Fig. 5. The correlation coefficient between predicted and measured abundance calculated for the LSCC data is $k = 0.91$. This value is an estimate of the prognosis quality. Overall this map demonstrates realistic distribution of TiO_2 over the lunar surface, but the following problems exist: (1) the abundance for maria is

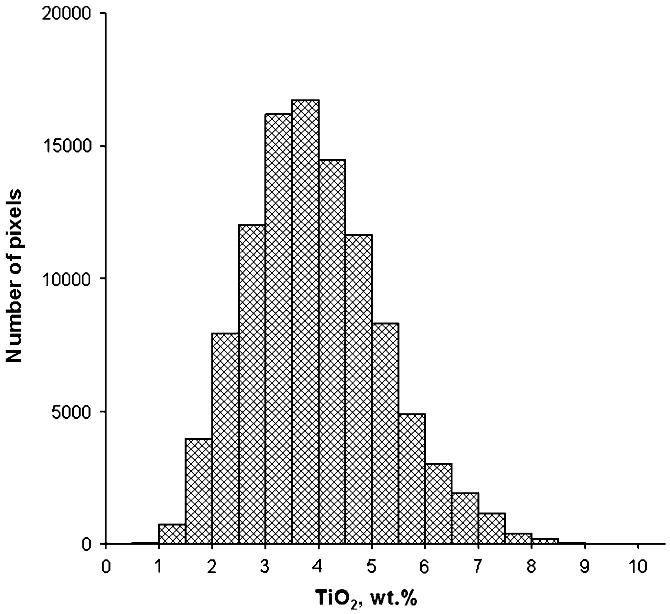

Fig. 6. Histogram of TiO_2 concentrations built for mare regions (with reflectance <0.14 at 750 nm band) from the MLR prognosis.

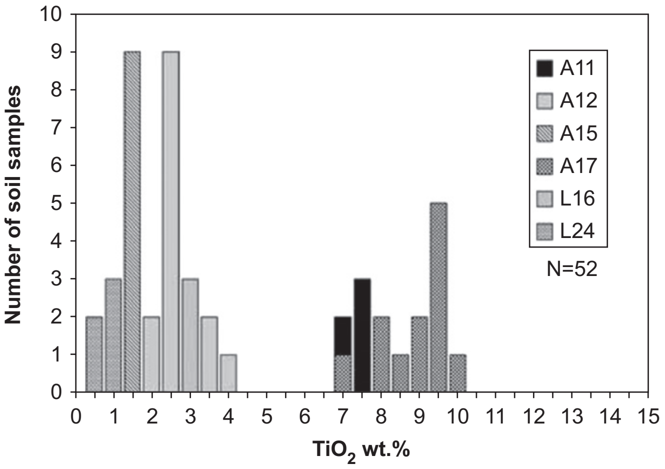

Fig. 7. Histogram of TiO_2 concentrations built for representative basaltic soils from individual Apollo (A) and Luna (L) sampling sites (Gillis et al., 2003).

too low (maximum 6%, while some LSCC samples give the values up to 10%); (2) the abundance for highlands is overestimated as compared with sample data; (3) the map contains significant noise. Statistical analysis of MLR prognosis map reveals a unimodal TiO_2 distribution for mare regions (see Fig. 6). Nearside mare basalts reveal a unimodal continuum of TiO_2 concentrations according to remote-sensing data (Giguere et al., 2000), whereas both basaltic rock samples and the corresponding soil samples (i.e., <1 mm regolith) from the lunar landing sites exhibit a bimodal distribution of TiO_2 concentrations (Neal and Taylor, 1992).

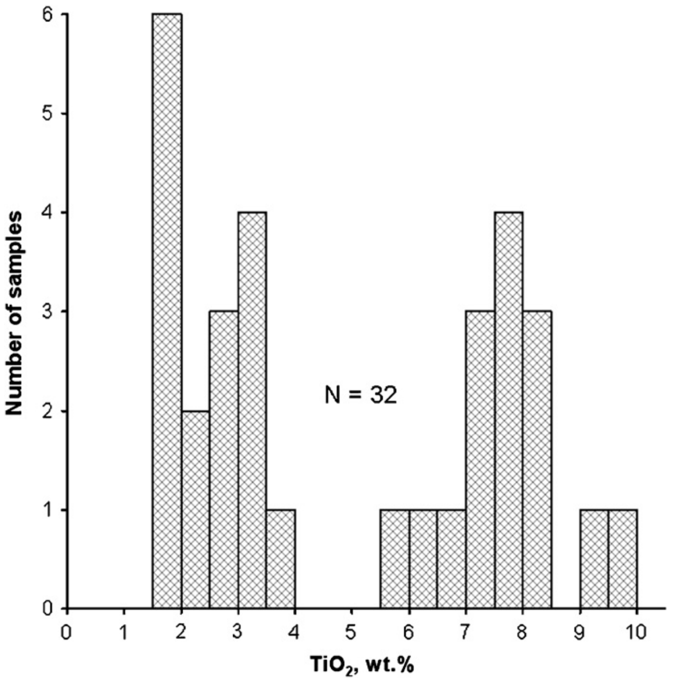

Fig. 8. Histogram of TiO_2 concentrations built for mare soils from LSCC data.

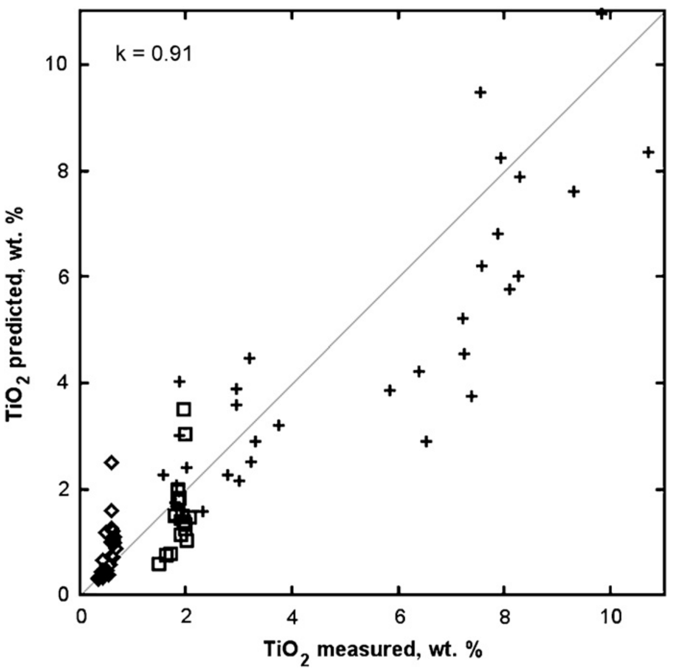

Fig. 9. Scatterplot for measured and MLR predicted (Eq. (3)) values for LSCC TiO_2 data. Crosses are mare samples, diamonds and rectangles are Apollo-14 and Apollo-16 samples, respectively.

A histogram made for representative basaltic soils from individual Apollo and Luna sampling sites is shown in Fig. 7. We have constructed an analogous histogram for mare samples from LSCC data used in (Shkuratov et al., 2005a; Pieters et al., 2006). Totally 8 samples × 4 size

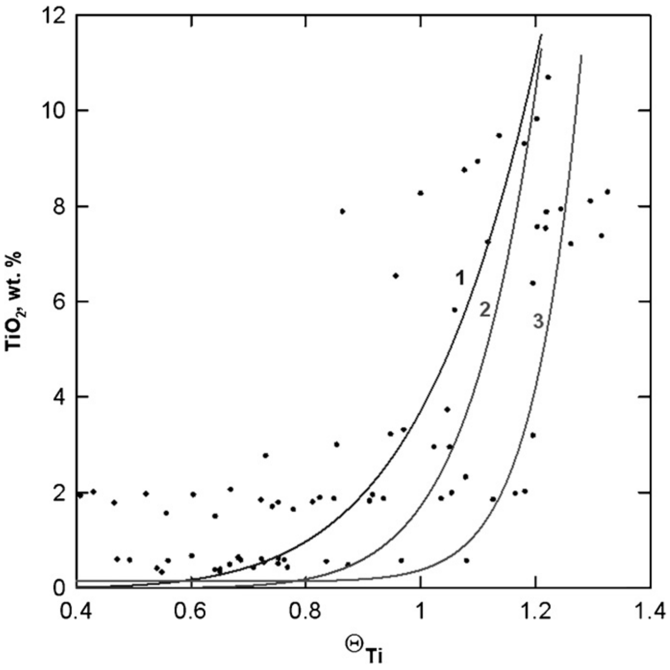

Fig. 10. TiO_2 abundance versus the spectral titanium parameter Θ_{Ti} for laboratory measurements of LSCC samples (dots): (1) presents Lucey's et al. prognosis (Eq. (2)), (2) and (3) correspond to Gillis' et al. regressions.

fractions were used (Fig. 8). Both histograms show a gap between 4% and 6% of TiO_2 content, which is not the case for MLR prognosis (cf. Fig. 6).

This discrepancy suggests several possibilities: the bimodal distribution of TiO_2 in the sample collections is an artifact of sampling, as was noted by Gillis et al. (2003), or the assumption on linearity of the relationships between optical and composition parameters is not adequate. The last assertion is confirmed by the character of correlation diagram (see scatterplot in Fig. 9): the plots for low-TiO_2 and high-TiO_2 mare samples, Apollo-14 and Apollo-16 highland samples do not lie on the same line and form separated groups. To point out the problem of compatibility of the orbital and samples data we have constructed the plot of TiO_2 abundance versus the spectral TiO_2 parameter Θ_{Ti} derived from laboratory measurements of LSCC samples (Fig. 10). As can be seen, the situation with approximation is not so optimistic, as in Figs. 1 and 4. The individual samples on the diagram are more scattered near regression lines than even in MLR case (cf. Fig. 9). We should note that spectral effect of maturity degree and particle size potentially limits the reliability of TiO_2 mapping techniques.

We conclude that the TiO_2 prognosis methods discussed above either do not adequately reproduce the statistical distribution of lunar sample data or not follow spectrochemical trends formed by groups of LSCC samples. We study this problem using an alternative non-linear method for the TiO_2 prognosis based on laboratory measurements of LSCC samples.

2. Preparation of LSCC and CSR data

2.1. LSCC data

Spectral and composition data for lunar soils used in this paper are summarized in Table 1 and discussed below. For analysis we use a suite of lunar samples by Lunar Soil Characterization Consortium (LSCC). This suite includes mare as well as highland soils (Taylor et al., 2001, 2003). The LSCC suite represents both compositional diversity of lunar soils and wide variations of their maturity degree that is characterized with the parameter I_s/FeO (Morris, 1976,1978). It is well known that the optical properties of lunar soils are controlled by small-size fractions (Pieters, 1993) and space weathering processes (Pieters et al., 2000). The LSCC collection we used includes the $<45\,\mu m$ components of lunar soils as those corresponding to optical properties measured by remote-sensing techniques. Each soil sample was subdivided into three size fractions (<10, 10–20, and 20–45 μm) in addition to "bulk" $<45\,\mu m$ samples (see Taylor et al., 2001, for description). High-resolution bi-directional spectra were acquired for each subsample with the RELAB spectrometer at the Brown University (Pieters and Hiroi, 2004). These coordinated spectral and chemical measurements provided for each sample give a possibility to account for changes of particle size as well as maturity degree (as correlated with particle size) in TiO_2 prediction. All spectra were measured over the spectral range 0.3–2.6 μm with 5 nm spectral resolution at a phase angle of 30°, see details in (Pieters and Hiroi, 2004). The LSCC data (spectra and composition) are available: http://www.planetary.brown.edu/relabdocs/LSCCsoil.html. The optical data listed in Table 1 are result of resampling the high-resolution RELAB spectra to the five CSR UV–VIS bands.

2.2. Refinement of CSR data with PCA

We used the principal component analysis (PCA) of 1-km CSR UV–VIS mosaics of reflectances ($\lambda = 415$, 750, 900, 950, and 1000 nm) (Eliason et al., 1999) reducing their resolution to 8-km resolution. The PCA allowed us to refine these mosaics for further analysis.

As is known (e.g., Jaumann, 1991), PCA gives a system of eigenfunctions, $\varphi_q(\lambda)$, that allow the Clementine UV–VIS image cube to be described as a series with a minimal number of terms:

$$A(x,y,\lambda) \equiv \bar{A}(\lambda) + \sum_{q=1}^{5} c(x,y)_q \varphi_q(\lambda), \tag{4}$$

where $c(x, y)_q$ are the principal components, $\bar{A}(\lambda)$ is the average image of the whole Clementine UV–VIS dataset, and x and y are coordinates of the pixel of images. We found that the fourth and fifth components are very noisy and do not present detectable spectral variations. We reject these components replacing them by their mean values and return to the reflectance presentation from the rest three principal components. This allows us to reduce random and regular

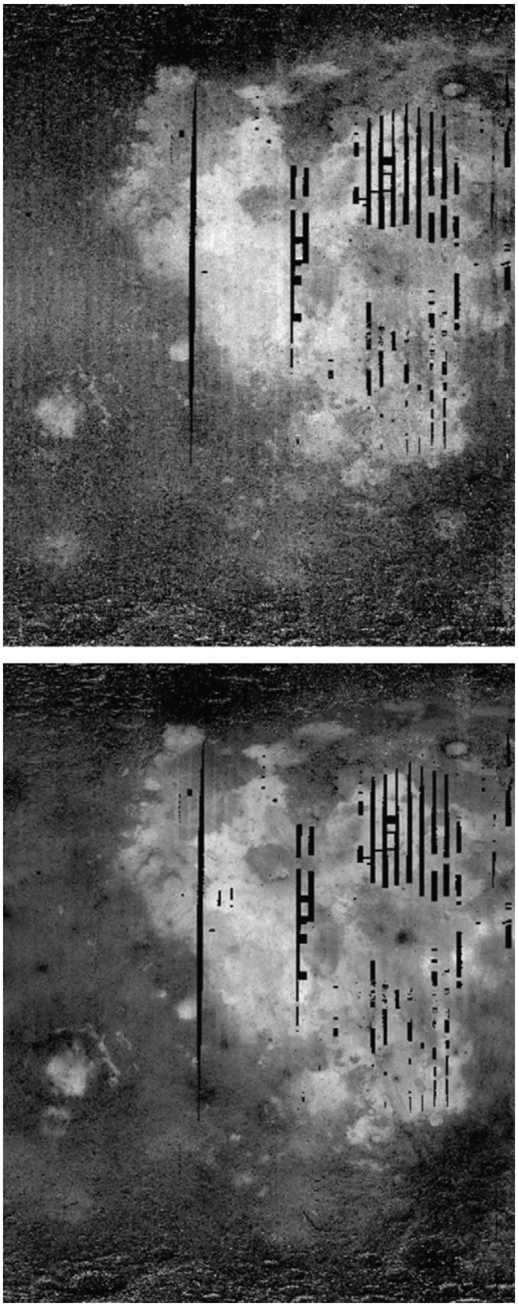

Fig. 11. Maps of TiO_2 abundance prognosis using MLR: before PCA filtration (upper panel), after PCA filtration (lower panel).

(e.g., the latitude trend) errors of the Clementine mosaics. In Fig. 11 we show a TiO_2 abundance map with the MLR algorithm without (a) and with (b) the PCA filtration. We also immediately note a drop of overall noisiness of the TiO_2 map and less visible vertical strips on the map. So the result looks more spatially coherent, as would be expected on the lunar surface at such a spatial resolution.

3. Artificial neural networks (ANNs)

In Section 1, we have discussed several methods to establish the best mathematical link between spectral and chemical data. For this aim one needs to search for the closest correlation between TiO_2 abundance and an empirical combination of spectral albedo of lunar samples in the CSR bands. In our previous works (Shkuratov et al., 2003a, 2005b) we tested several empirical formulas minimizing the RMS error between the observed and the predicted values of TiO_2 content. This approach particularly depends on the choice of empirical combinations of spectral bands, thus the general weakness of this technique is that there is no unique form of relationship connecting spectral and chemical parameters. When the number of spectral bands are limited (CSR bands), the spectral resolution is insufficient for reliable identification and characterization of individual electronic transition mineral absorption bands. Insufficient spectral resolution in principle cannot give a unique solution for spectro-chemical relationships, however, there is a technique suggesting the "maximally effective" way to analyze relationships between bands-restricted spectral data and chemical/mineral composition. This is the technique of ANNs. An interesting and useful feature of the technique is that it does not need to specify the formulae for finding the statistical relationships, examples of which are Eqs. (1)–(3).

There are many sorts of neural networks for different applications. However the multilayer perceptron (Rumelhart and McClelland, 1986) with the teaching method of errors back propagation (Haykin, 1994) is most suitable for regression tasks. Typical architecture of a multilayer perceptron is shown in Fig. 12. An ANN consists of several layers of neurons. Each neuron (Fig. 13) has several inputs and one output. A neuron in a previous layer is connected with all neurons in the next layer. Output value of a neuron is calculated using the following equation:

$$\mathrm{Out} = f\left(\sum x_i \times w_i\right), \qquad (5)$$

where x_i is the value and w_i is the adjusted weight on ith input, f is a non-linear activation function. Usually the so-called "sigmoid" function is used

$$f(s) = 1/(1 + \mathrm{e}^{-\alpha \times s}), \qquad (6)$$

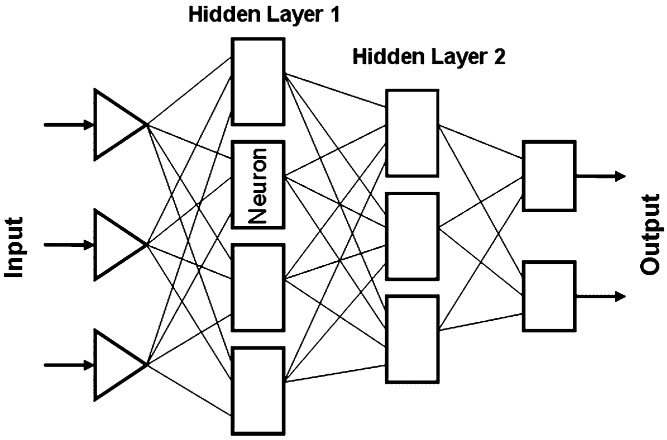

Fig. 12. Typical architecture of a multilayer perceptron.

where α is the parameter of non-linearity. The output value(s) of whole ANN are produced by consecutive applying of Eq. (5) to all neurons in all layers. Using of multilayer perceptrons for solving tasks of multidimensional non-linear regression includes two stages:

Teaching (training): Using a set of patterns (a set of input and corresponding output values) the ANN adjusts neurons weights w_i to minimize errors between computed output values and values from the patterns. During this process, randomly selected patterns from the whole teaching dataset are presented many times to the ANN recursively adjusting neuron weights w_i. Presentation of all patterns from teaching dataset is called "epoch of teaching". Usually several hundreds or thousands epochs are used for training that often requires significant computer resources. Before using for training, the patterns dataset must be standardized: input and output values are reduced to $-1.0\ldots+1.0$ range:

$$X_i = (x_i - x_{0i})/xk_i \tag{7}$$

and to 0.0 … 1.0 range:

$$Y_i = (y_i - y_{0i})/yk_i, \tag{8}$$

respectively, where x_{0i} and xk_i are the standardizing parameters for inputs, and y_{0i} and yk_i are the standardizing parameters for outputs. As has been mentioned we used six spectral parameters (see Eq. (3)): A_R, C_{BR}, C_{IR1}, C_{IR2}, C_{IR3}, and D. The standardizing parameters for LSCC data (Table 1) are given in Table 2.

Computation (prediction): The taught ANN computes a prediction using Eq. (5) and (6) on the base of experimental input values x_i standardized in the same way as input values of the training patterns. Because of ANN gives an output value in 0.0 … 1.0 range, predictions are corrected on corresponding coefficients:

$$y_i = yk_i Y_i + y_{0i}. \tag{9}$$

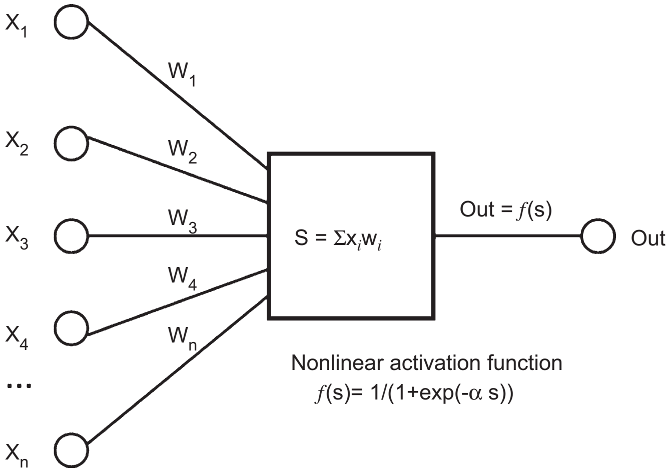

Fig. 13. Artificial neuron.

Table 2
Standardizing parameters for ANNs inputs

	A_R	C_{BR}	C_{IR1}	C_{IR2}	C_{IR3}	D	SC_{A16}	SC_{A14}	$SC_{M\ low\text{-}Ti}$	$SC_{M\ high\text{-}Ti}$
x_0	0.2431	0.6288	0.9832	0.9996	1.0402	1.1336	0.5000	0.5000	0.5000	0.5000
x_k	5.6007	6.5615	5.3248	4.5291	4.3326	4.4592	2.0000	2.0000	2.0000	2.0000

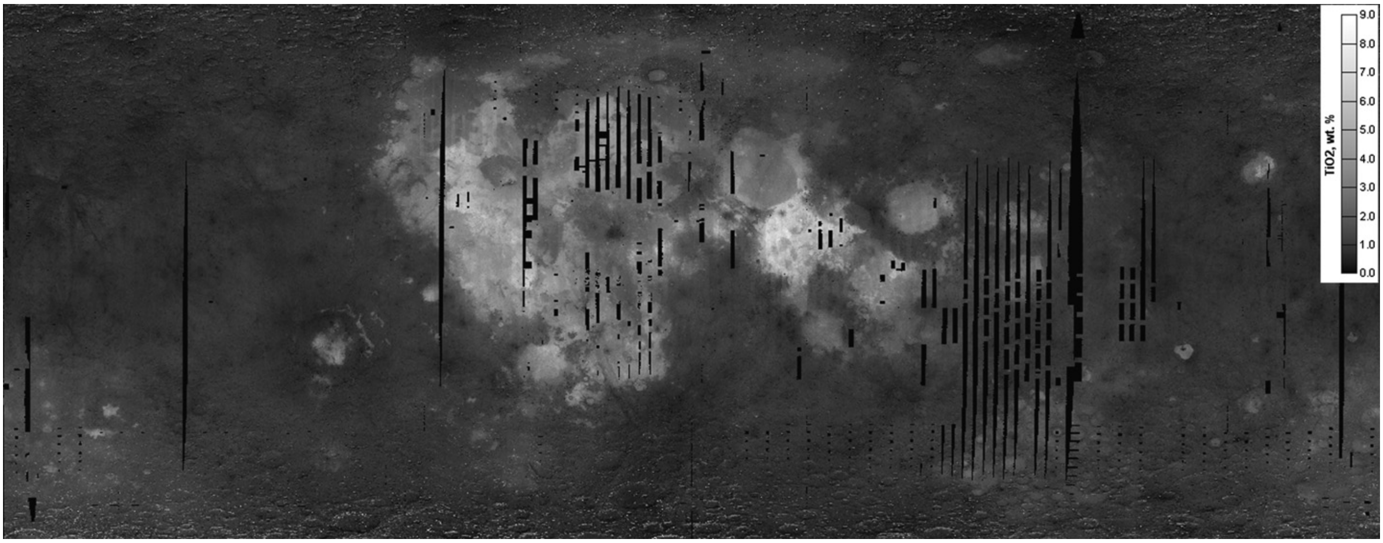

Fig. 14. Map of TiO_2 abundance using moderately non-linear ANN.

For TiO_2, $y_{0i} = -0.178$ and $yk_i = 11.396$. With this correction taught ANN computes output values very rapidly. Finally, we also note that in contrast to linear regressions ANNs can calculate prediction of several output values simultaneously.

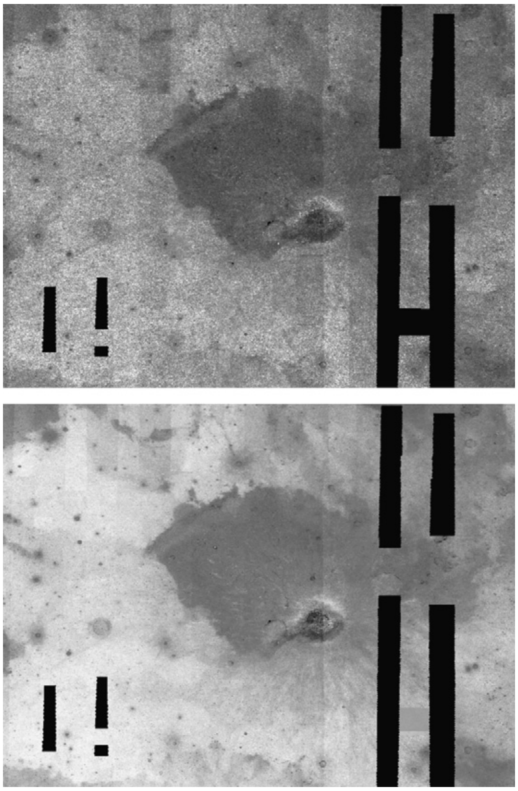

Fig. 15. Map of TiO_2 abundance with 1 km resolution: MLR (upper panel), moderately non-linear ANN (lower panel). Aristarchus Plateau is seen near the center. TiO_2 scale is the same as in Fig. 14.

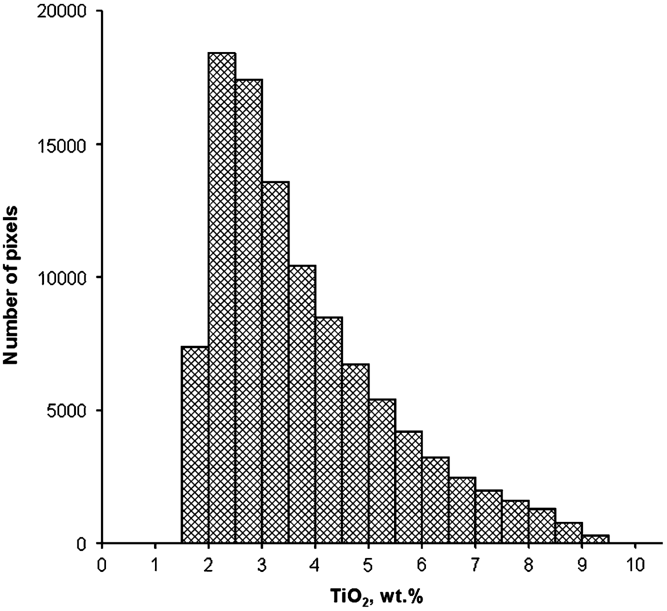

Fig. 16. The same as in Fig. 6, but for a moderately non-linear ANN prognosis.

4. TiO_2 prognosis using artificial neural networks

For reliability of the ANNs prognosis an adequate architecture and parameters of the ANN should be carefully chosen. Too non-linear ANN is subjected to so-called "overtraining", when the ANN is trained not only by "useful" information in training patterns. This may result in "supersaturation" for minimal and maximal predicted values and in appearance of significant noise in the prognosis. On the other hand, an insufficiently non-linear ANN will not be able to track all regularities in

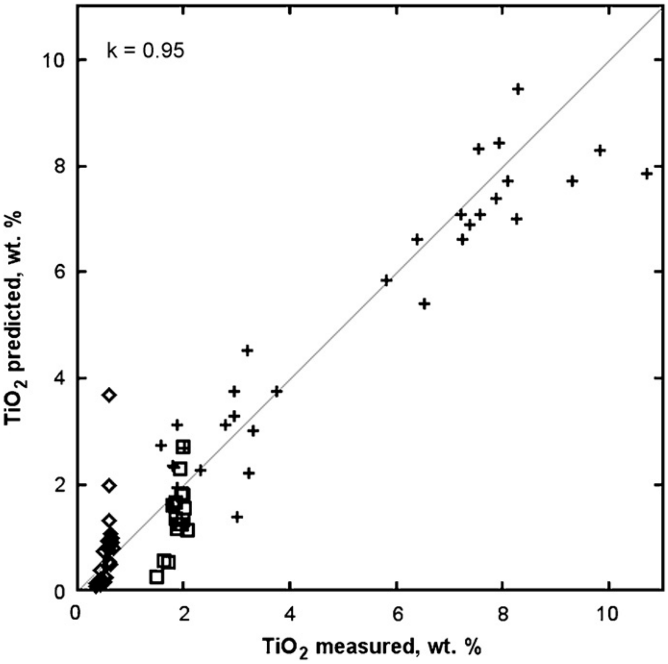

Fig. 17. Scatterplot for measured and ANN predicted values for TiO_2 (crosses are mare samples, diamonds and rectangles are Apollo-14 and Apollo-16 samples, respectively). A moderately non-linear ANN is used.

Table 3
Input weights of neurons in first hidden layer of TiO_2 prognosis ANN

N_i	A_R	C_{BR}	C_{IR1}	C_{IR2}	C_{IR3}	D
0	0.4986	2.3223	0.5409	1.2126	−0.1447	−0.0355
1	0.2062	1.8351	0.9909	−0.2891	−1.0505	−0.6925
2	1.5316	1.6740	−0.7028	0.3696	1.6691	1.2761
3	0.7294	−3.3990	−1.8817	−0.5255	1.2135	4.7384

N_i is neuron index.

Table 4
Input weights of neurons in second hidden layer of TiO_2 prognosis ANN

N_i	w_0	w_1	w_2	w_3
0	1.4344	−1.2085	3.2410	5.3733
1	2.0864	−1.0830	0.0737	−0.8752

N_i is neuron index.

training dataset. It is very important to define criteria of optimality of the ANN architecture and parameters. Such criteria may be: (1) maximal correlation between measured and calculated values for training dataset; (2) similarity of statistical characteristics (e.g. histogram) of training data (LSCC data in our case) and predicted data (prognosis with CSR data in our case).

Table 5
Input weights of neurons in output layer of TiO_2 prognosis ANN

Neuron	w_0	w_1
TiO_2	−2.8466	−2.3900

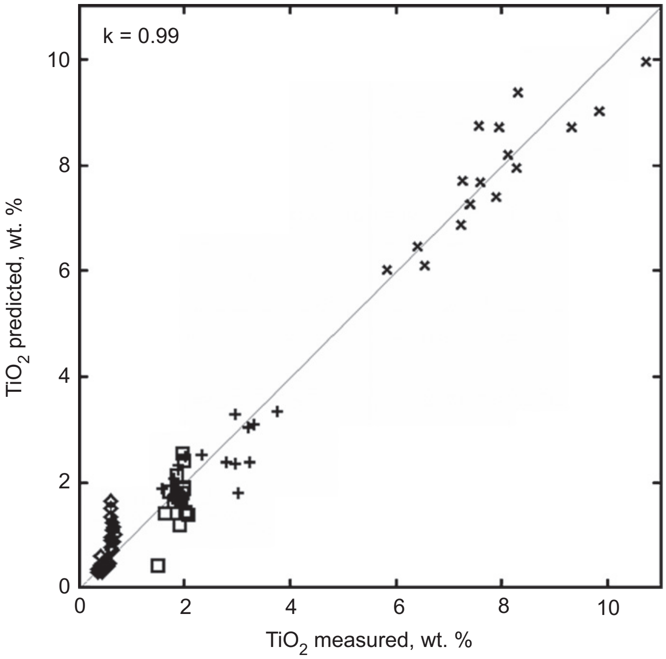

Fig. 18. The same as in Fig. 17, but a significantly non-linear ANN is used.

4.1. Using simple ANN

Initially we use a perceptron (Rumelhart and McClelland, 1986) with one hidden layer and with 3 neurons for studying the links between the spectral parameters a_i and composition parameter TiO_2 (Eq. (3)). The input layer contains the number of neurons equal to the spectral parameters (6 in our case), the output layer has one neuron. This structure and parameter $\alpha = 1.0$ of the ANN are chosen to make the ANNs moderately non-linear. The

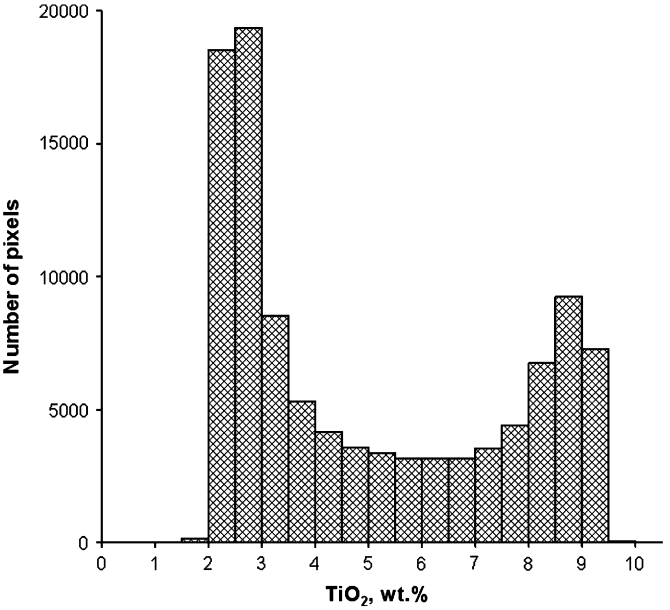

Fig. 20. The same as in Fig. 6, but for significantly non-linear ANN prognosis.

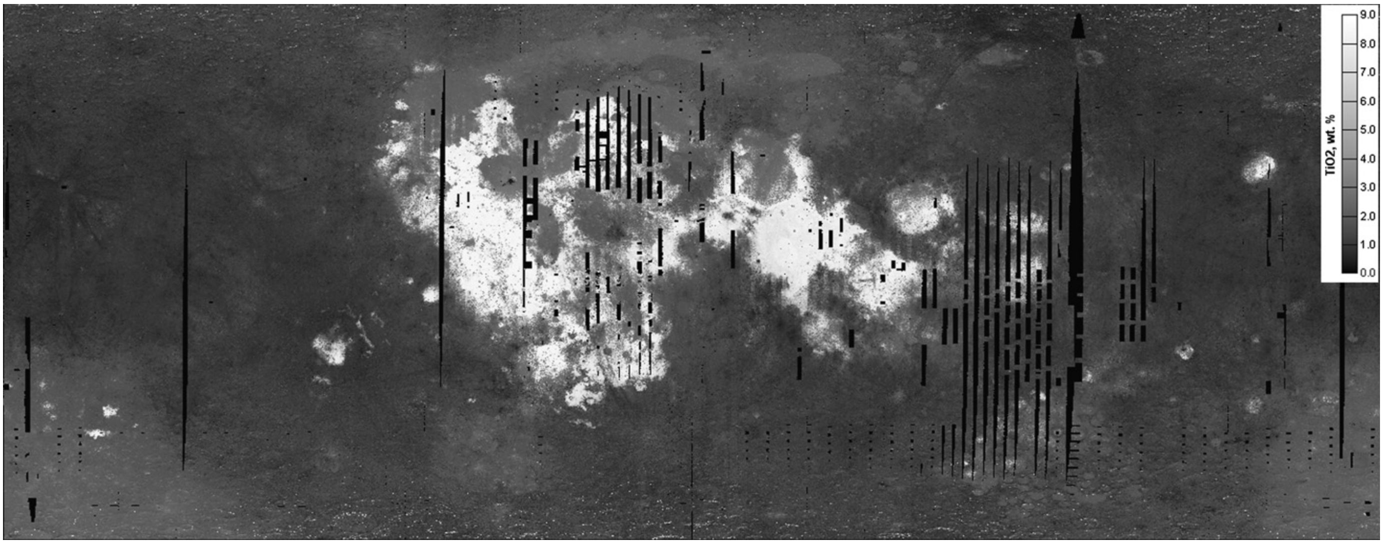

Fig. 19. Map of TiO_2 abundance using significantly non-linear ANN.

ANN is trained by LSCC dataset (Table 1) during 500 epochs.

Result of using such an ANN for the TiO_2 prognosis is shown in Fig. 14. As compared with the MLR prediction this new result gives higher correlation coefficient between predicted and measured abundance calculated for the LSCC data ($k = 0.95$ in contrast to $k = 0.91$ for MLR). Maximal values of TiO_2 abundance increased from 6% for MLR to 7.5%, minimal values became lower (0.2%) that is more realistic, spatial noise significantly decreased (cf. Fig. 15a and b). Nevertheless the histogram of distribution of TiO_2 concentrations for mare regions is still unimodal (Fig. 16), i.e., the second criterion of optimality of the ANN architecture is not kept. The analysis of the correlation diagram (Fig. 17) shows that such an ANN is not satisfactory reproduces four groups of samples (Apollo-16, Apollo-14, low-TiO_2, and high-TiO_2) corresponding to different classes of the lunar surface.

Therefore we decided to use a significantly non-linear ANN (2 hidden layers, 4 neurons in the first layer, 2 neurons in the second layer, $\alpha = 1.5$, training epochs number = 4000). Adjusted neurons weights are presented in Tables 3–5. Corresponding correlation diagram for LSCC data is shown in Fig. 18. Correlation between predicted and measured TiO_2 values became significantly higher ($k = 0.99$) and points for all kinds of samples lie close to the diagonal line. Results of prognosis with CSR data is shown in Fig. 19. Distribution of TiO_2 over the lunar surface for mare regions became bimodal (Fig. 20) and is very similar to histograms obtained from LSCC sample collections (cf. Fig. 4). At the same time the second maximum of the histogram is some greater (8.5%) in comparison with the LSCC sample data (7.5%). Adjusting the ANN to be less non-linear shifts the maximum to small values, but the maximum level becomes too low, with gradual disappearing the second high-TiO_2 mode. Furthermore, some Apollo-16 points are still out of the major trend along the diagonal on the correlation diagram (Fig. 18). It may cause errors in prognosis for highland regions. Thus, in spite of evident progress, using the simple ANN does not allow us to solve the mentioned problems of the TiO_2 prognosis.

4.2. ANN prognosis with spectral classification

To take into account different behavior of correlation links between optical and compositional parameters for different surface areas (or optical types) we use a system of two ANN. The first one ("Classification ANN") is used for a spectral classification of the points on the CSR mosaics. The second one ("Prognosis ANN" with 10 inputs and 1 output) produces a prediction using optical parameters directly from CSR data and information about spectral classes from the first ANN. The "Classification ANN" has 6 inputs and 4 outputs and provides the so-called "soft" classification: for each pixel in CSR mosaics the 4 parameters $SC_{M\ \text{low-Ti}}$, $SC_{M\ \text{high-Ti}}$, SC_{A14}, and SC_{A16} (corresponding to low-Ti mare, high-Ti mare, Apollo-14, and Apollo-16 soils) in the 0.0 … 1.0 range are calculated. Both ANNs have the same structure: 1 hidden layer with 3 neurons ($\alpha = 1.2$). "Classification ANN" and "Prognosis ANN" are trained during 500 and 1000 epochs, respectively, using LSCC dataset (Table 1). Adjusted neurons weights are presented in Tables 6–9.

Table 6
Input weights of neurons in hidden layer of classification ANN

N_i	A_R	C_{BR}	C_{IR1}	C_{IR2}	C_{IR3}	D
0	4.3693	2.0469	1.1328	1.7261	0.0593	−1.2040
1	1.1415	−4.6267	−1.2505	−1.1236	0.4143	3.5966
2	2.5235	0.9834	−0.2454	0.5972	2.1901	1.5583

N_i is neuron index.

Table 7
Input weights of neurons in output layer of classification ANN

Neuron	w_0	w_1	w_3
SC_{A16}	3.1614	−0.3404	2.0586
SC_{A14}	−1.4904	2.4493	2.2257
$SC_{M\ \text{low-Ti}}$	−2.4729	2.6110	−2.8712
$SC_{M\ \text{high-Ti}}$	−3.0804	−5.5366	−1.3075

Table 9
Input weights of neurons in output layer of TiO_2 prognosis ANN

Neuron	w_0	w_1	w_2
TiO_2	1.0080	−2.0689	1.5649

Table 8
Input weights of neurons in hidden layer of TiO_2 prognosis ANN

N_i	A_R	C_{BR}	C_{IR1}	C_{IR2}	C_{IR3}	D	SC_{A16}	SC_{A14}	$SC_{M\ \text{low-Ti}}$	$SC_{M\ \text{high-Ti}}$
0	0.4536	0.7971	0.6376	0.3895	−0.3593	−0.0899	−0.6776	0.0313	0.6993	1.6563
1	−0.0529	1.0228	0.0240	1.1849	1.0340	0.6085	0.5118	0.3060	0.1295	−0.5089
2	−0.5791	1.0474	0.6103	0.4715	0.6090	0.1860	−0.7571	0.4387	0.0831	1.7581

N_i is neuron index.

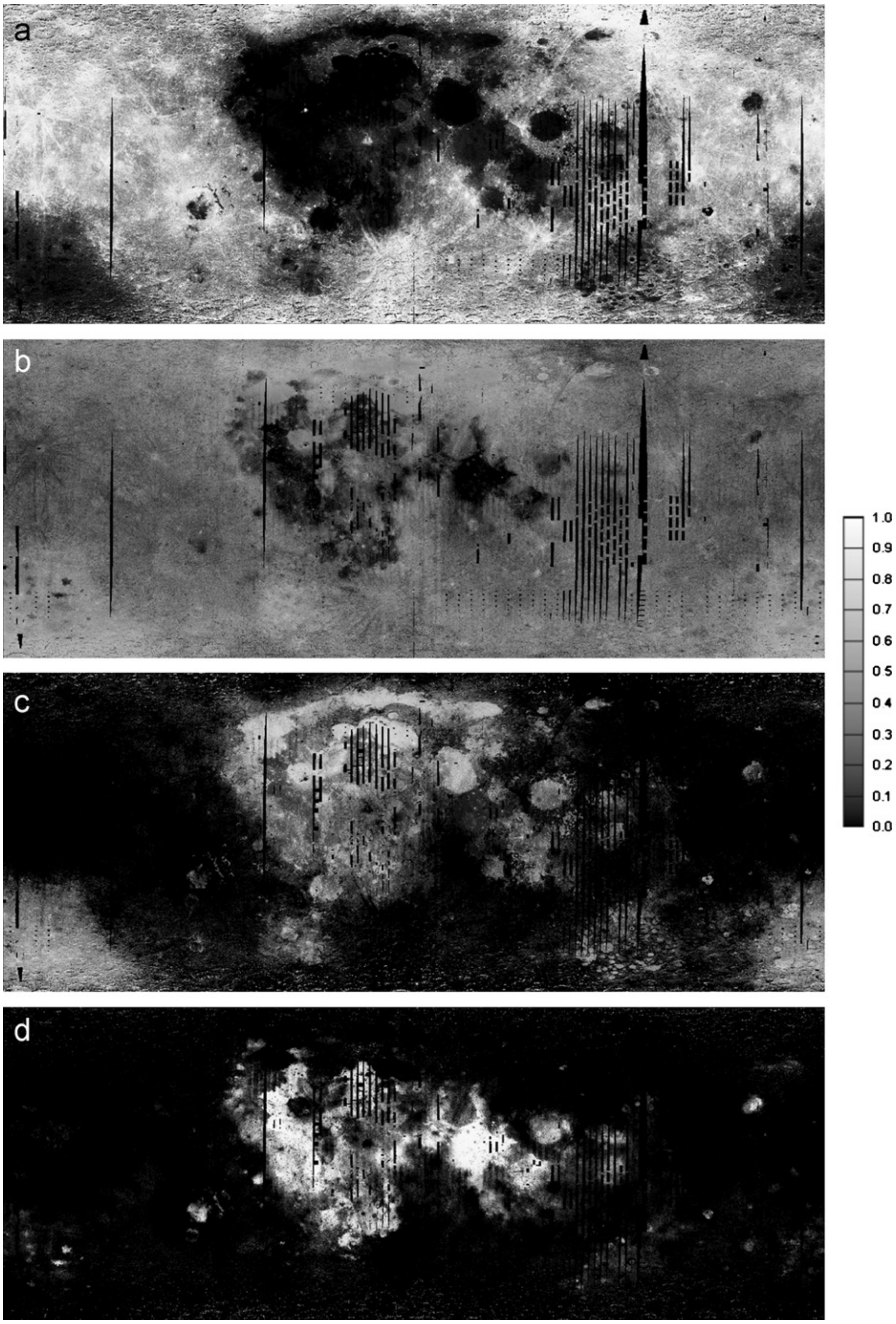

Fig. 21. Spectral classification of the lunar surface by the similarity to the LSCC samples: (a) Apollo-16, (b) Apollo-14, (c) low TiO_2 maria, (d) high TiO_2 maria. Brighter tones correspond to higher similarity.

In Figs. 21–24 we show results of this prognosis system. A spectral classification of the lunar surface by similarity to the LSCC samples using the classification ANN is presented in Fig. 21. Values of the similarity parameter <0.5 correspond to non-classified regions in terms of spectral classes. For example, maximal values shown on the Fig. 21b are ~0.5. This means that most of both highlands and mare areas cannot be classified as "spectrally Apollo-14-like areas". Correlation diagram for LSCC data (Fig. 22) shows that points corresponding to different classes fit rather close to the same diagonal line. The Apollo-16 group became very compact, which gives confidence in correctness of the TiO_2 prognosis for Apollo-16-like highland areas. The obtained distribution (Fig. 23) is very similar to that built with the previous method (Fig. 19), excluding regions with lowest and highest values of TiO_2. The histogram of TiO_2 distribution (Fig. 24) is now very similar to that in Fig. 7, i.e., all criteria of the optimality of the ANN architecture are satisfied.

These new results provided by a system of two weakly non-linear ANNs lead to conclusion that the relations between optical parameters and abundance of TiO_2 inside each spectral class is nearly linear, but the character of these relations is very different from one optical class to others.

Thus, we can conclude that using the system of two ANNs allowed resolving several problems of the TiO_2 prognosis. In the Table 10 the results of the ANN prediction for regions of Apollo and Luna stations landing sites are compared with other approaches (linear regression and Lusey's approach). Values of TiO_2 content in the table were calculated by averaging 3×3 pixels areas on 8 km maps. In general results are in good agreement, but, as was mentioned above, the MLR approach gives too high values for highlands (Apollo 16) and Lucey's approach shows a very high maximal value for maria (Apollo 11). The reference bulk values for landing sites from Lunar Sourcebook (Heiken et al., 1991) are also presented in Table 1 (for mare basaltic samples, soil, and regolith breccias, and also polymict breccias). We note that all prognoses are very bad for the landing site of Luna 24 in Mare Crisium.

Another prognosis of TiO_2 in lunar soils is based on Gillis' et al. (2003) approach. Their map (see Fig. 14 in (Gillis et al., 2003)) reveals the distribution of TiO_2 in the maria to be similar to our map, but one should keep in mind that Gillis' et al. method is applicable for mare regions only.

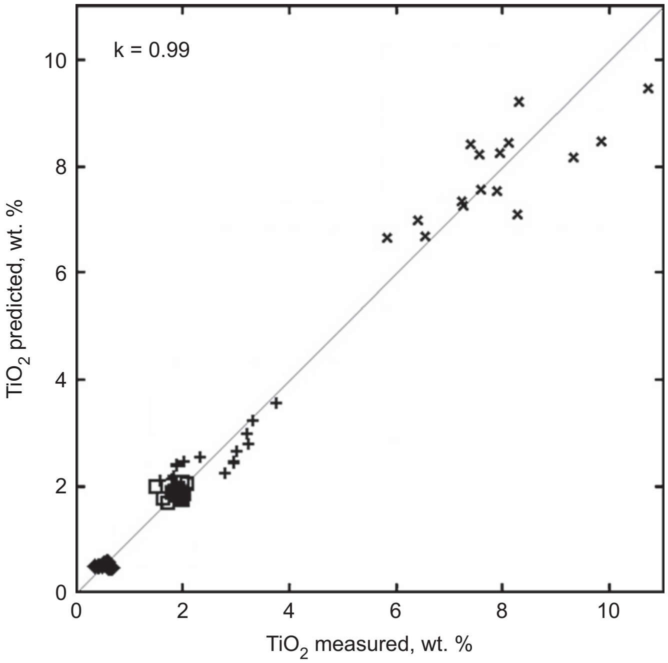

Fig. 22. Scatterplot for measured and ANN predicted values for TiO_2 (crosses are mare samples, diamonds and rectangles are Apollo-14 and Apollo-16 samples, respectively). Prognosis with spectral classification is used.

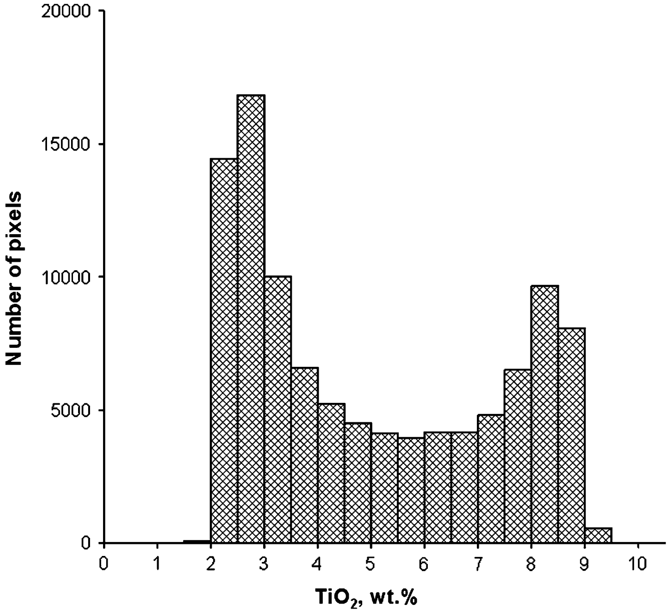

Fig. 24. The same as in Fig. 6, but for the ANN prognosis with spectral classification.

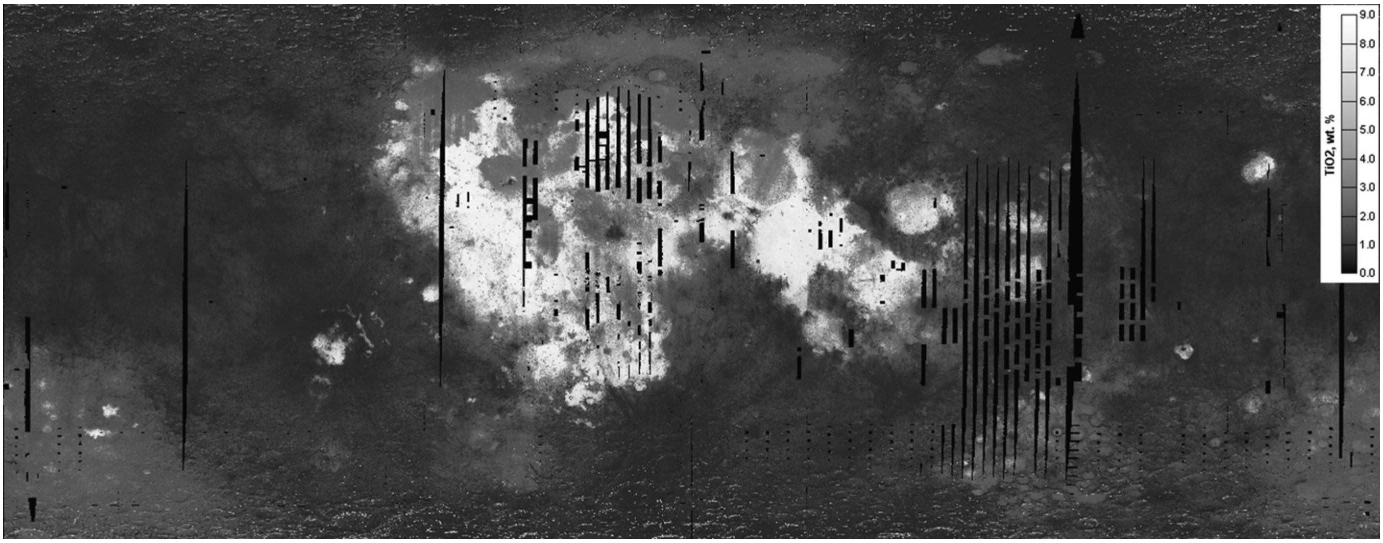

Fig. 23. Map of TiO_2 abundance ANN prognosis with spectral classification.

Table 10
Comparison of prediction of TiO_2 abundance (wt%) using our approach (ANN) with other prediction approaches (linear regression (MLR), Lucey's and Gillis's (Gillis et al., 2003, Table 1)) and measurements of returned samples (LSCC—average LSCC data values; SKO—average values from (Shkuratov et al., 1999, Table 1); LSB—data from Lunar source-book (Heiken et al., 1991), Table 8.1: MBAS = mare basalts, S&RB = soils and regolith breccias and BX = polymict breccias

Site	Longitude (deg)	Latitude (deg)	Predicted TiO_2 abundance (wt%)				Measured TiO_2 abundance (wt%)				
			ANN	MLR	Lucey's	Gillis's	LSCC	SKO	LSB (MBAS)	LSB (S&RB)	LSB (BX)
Apollo 11	23.5	0.8	7.9±0.2	6.2±0.1	11.7±0.7	7.1	7.76±0.46	7.40±1.85	10.41±1.02	7.94±0.62	–
Apollo 12	−23.2	−2.9	6.9±0.6	5.1±0.1	6.4±0.5	3.5	3.15±0.30	2.68±0.27	3.62±0.87	2.61±0.10	–
Apollo 14	−17.4	−3.7	2.2±0.3	3.2±0.2	1.8±0.3	–	1.88±0.15	1.72±0.34	–	–	1.51±0.23
Apollo 15	3.7	25.8	2.0±0.7	3.1±0.5	1.7±0.6	1.2	1.89±0.22	1.64±0.49	1.97±0.39	1.46±0.34	1.68±0.32
Apollo 16	15.1	−9.0	0.8±0.1	2.3±0.2	0.8±0.1	–	0.53±0.10	0.55	–	0.53±0.14	0.64±0.45
Apollo 17	30.7	20.1	4.8±2.1	3.7±1.8	3.7±1.8	10.0	7.92±1.45	9.50	11.94±1.11	2.84±2.01	2.23±2.47
Luna 16	56.5	−0.8	7.2±0.4	5.1±0.2	6.4±0.6	2.9	–	3.36±0.34	4.77±0.49	3.30±0.10	–
Luna 20	56.5	3.5	1.7±0.2	2.9±0.1	1.3±0.1	–	–	0.47±0.09	–	0.50±0.06	0.49±0.22
Luna 24	52.1	12.7	5.2±0.9	4.7±0.2	5.0±0.5	1.6	–	1.15±0.23	1.02±0.19	1.04±0.15	–

Values and standard deviations for ANN, MLR and Lucey's have been calculated for 3×3 pixels areas on 8-km maps.

According to the new TiO_2 map in Fig. 23 broad mare areas in Procellarum and Mare Imbrium and also the whole Mare Tranquillitatis show maximal TiO_2 content on the lunar surface (~8.5–9% at 8-km resolution). Northern maria, like Frigoris and NW part of Procellarum, belong to the low-Ti mare subclass with TiO_2 content ~3–4 wt%. We also confirm the compositional difference of Mare Serenitatis and Mare Tranquillitatis by the TiO_2 content. The South pole—Aitken basin clearly shows up on the map with intermediate 5–6% TiO_2 content. The TiO_2 content for highland regions varies in the $<2\%$ range.

Spectral characteristics of the lunar soils are significantly affected by particle size variation and soil maturity as well as the iron content. These factors influence the results of prognosis and were taken it into account empirically from the LSCC data during teaching the ANNs. Features of the ANN technique allow us to be sure that the information is used with maximal effectiveness in comparison with traditional linear regressions. If any additional information about relationships between spectral characteristics and composition of lunar soils will be accessible, it may be used for teaching the ANNs improving the quality of prognosis.

5. Conclusions

Algorithms and software based on artificial neural networks (ANN) technique have been developed for prognosis of chemical and mineralogical composition of the lunar surface using Clementine UV–VIS maps and LSCC data.

Using the ANN technique allows us to study empirical links between spectral characteristics of lunar soils and composition parameters without any restrictions on the character of these relations.

A new non-linear method for prognosis of TiO_2 abundance based on the ANN approach is proposed and a new map of TiO_2 distribution over the lunar surface is constructed. The method reliably reproduces the statistical distribution of LSCC data for mare basalts.

Our results could be useful for the strategy in analysis of lunar data obtained with spacecrafts especially for Chandrayaan-1 and Lunar Reconnaissance Orbiter missions.

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References

Adams, J.B., McCord, T.B., 1971. Alteration of lunar optical properties: age and composition effect. Science 171 (3971), 567–571.

Adams, J.B., McCord, T.B., 1972. Electronic spectra of pyroxenes and interpretation of telescopic spectra reflectivity curves of the Moon. In: Proceedings of the Third Lunar Science Conference, LPI, Houston, TX.

Blewett, D.T., Lucey, P.G., Hawke, B.R., Joliff, B.L., 1997. Clementine images of the lunar sample-return stations: refinement of FeO and TiO_2 mapping techniques. J. Geophys. Res. 102, 16,319–16,325.

Burns, R.G., 1993. Mineralogical Application of Crystal Field Theory. Cambridge University Press, Cambridge, MA, 551pp.

Charette, M., McCord, T., Pieters, C., Adams, J., 1974. Application of remote spectral reflectance measurements to lunar geology classification and determination of titanium content of lunar soils. J. Geophys. Res. 79, 1605–1613.

Eliason, E.M., McEwen, A.S., Robinson, M.S., Lee, E.M., Becker, T., Gaddis, L., Weller, L.A., Isbell, C.E., Shinaman, J.R., Duxbury, T., Malaret, E., 1999. Digital processing for a global multispectral map of the Moon from the Clementine UVVIS imaging instrument. In: XXX Conference of Lunar and Planetary Science, Abstract #1933.

Elphic, R.C., Lawrence, D.J., Feldman, W.C., Barraclough, B.L., Gasnault, O.M., Maurice, S., Lucey, P.G., Blewett, D.T., Binder, A.B., 2002. Lunar Prospector neutron spectrometer constraints on TiO_2, J. J. Geophys. Res. 107 (E4), 5024.

Giguere, T.A., Taylor, G.J., Hawke, B.R., Lucey, P.G., 2000. The titanium contents of lunar mare basalts, Meteorit. Planet. Sci. 35 (1), 193–200.

Gillis, J., Jollif, B., Elphic, R., 2003. A revised algorithm for calculating TiO_2 from Clementine UV–VIS data: a synthesis of rock, soil, and remotely sensed TiO_2 concentrations. J. Geophys. Res. 108 (E2).

Hawke, B.R., Peterson, C.A., Blewett, D.T., Bussey, D.B.J., Lucey, P.G., Taylor, G.J., Spudis, P.D., 2003. Distribution and modes of occurrence of lunar anorthosite. J. Geophys. Res. 108 (E6), 5050.

Haykin, S., 1994. Neural Networks: A Comprehensive Foundation. Macmillan College Publishing, New York.

Heiken, G., Vaniman, D., French, B., 1991. Lunar Source-book. Cambridge University Press, New York, 736pp.

Jaumann, R., 1991. Spectral–chemical analysis of lunar surface materials. J. Geophys. Res. 96 (E5), 22,793–22,807.

Johnson, J.R., Larson, S.M., Singer, R.B., 1991. A reevaluation of spectral ratios for lunar mare TiO_2 mapping. Geophys. Res. Lett. 18 (11), 2153–2156.

Jolliff, B.L., 1999. Clementine UV–VIS multispectral data and the Apollo 17 landing site: what can we tell and how well? J. Geophys. Res. 104 (E6), 14,123–14,148.

Lucey, P.G., Blewett, D.T., Hawke, B.R., 1998. Mapping the FeO and TiO_2 content of the lunar surface with multispectral imagery. J. Geophys. Res. 103 (2), 3679–3699.

Lucey, P.G., Blewett, D.T., Bradley, L.L., 2000. Lunar iron and titanium abundance algorithm based on final processing of Clementine ultraviolet–visible images. J. Geophys. Res. 105, 20,297–20,305.

McCord, T.B., Clark, R.N., Hawke, B.R., McFadden, L.A., Owensby, P., Pieters, C.M., Adams, J.B., 1981. Moon: near infrared spectral reflectance, a first good look. J. Geophys. Res. 86 (B11), 10883–10892.

Morris, R., 1976. Surface exposure indices of lunar rocks: a comparative FMR study. Proc. Lunar Planet. Sci. 7, 315–335.

Morris, R., 1978. The surface exposure (maturity) of lunar soils: some concepts and Is/FeO compilation. Proc. Lunar Planet. Sci. 9, 2287–2297.

Neal, C.R., Taylor, L.A., 1992. Petrogenesis of mare basalts: a record of lunar volcanism. Geochim. Cosmochim. Acta 56, 2177–2211.

Nozette, S., and the Clementine Team, 1994. The Clementine Mission to the Moon: Scientific overview. Science 266, 1835–1839.

Pieters, C.M., 1978. Mare basalt types on the front side of the Moon: A summary of spectral reflectance data. In: Proceedings of the Ninth Lunar and Planetary Science Conference, LPI, Houston, TX, pp. 2825–2849.

Pieters, C.M., 1986. Composition of the lunar highland crust from near-infrared spectroscopy. Rev. Geophys. Space Phys. 24 (3), 557–578.

Pieters, C.M., 1993. Compositional diversity and stratigraphy of the lunar crust derived from reflectance spectroscopy. In: Pieters, C.M., Englert, P.A.J. (Eds.), Remote Geochemical Analysis: Elemental and Mineralogical Composition. Cambridge University Press, Cambridge, pp. 309–339.

Pieters, C.M., 1999. The Moon as a spectral calibration standard enabled by lunar samples: the Clementine example. In: Workshop on New Views of the Moon II: Understanding the Moon through the Integration of Diverse Datasets, LPI, Houston, TX, pp. 47–49.

Pieters, C., Hiroi, T., 2004. RELAB (Reflectance Experiment Laboratory): a NASA multispectral spectroscopy facility. In: 35th Lunar and Planetary Science Conference, #1720 (LPI Houston)

Pieters, C.M., Taylor, L.A., McKay, D.S., Wentworth, S., Morris, R.V., Keller, L.P., 2000. Spectral characterization of lunar mare soils. Lunar Planet. Sci. 31, #1865.

Pieters, C.M., Stankevich, D.G., Shkuratov, Yu.G., Taylor, L.A., 2002. Statistical analysis of the links among lunar mare soil mineralogy, chemistry and reflectance spectra. Icarus 155, 285–298.

Pieters, C.M., Shkuratov, Yu.G., Kaydash, V.G., Stankevich, D.G., Taylor, L., 2006. Lunar soil characterization consortium analyses: pyroxene and maturity estimates derived from Clementine data. Icarus 184, 83–101.

Rumelhart, D.E., McClelland, J.L., 1986. Parallel Distributed Processing: Explorations in the Microstructure of Cognition. MIT Press, Cambridge, MA (v. 1: Foundations, 547pp.).

Shkuratov, Yu.G., Kaydash, V.G., Opanasenko, N.V., 1999. Iron and titanium abundance and maturity degree distribution on the lunar nearside. Icarus 137, 222–234.

Shkuratov, Yu.G., Kaydash, V.G., Kreslavsky, M.A., Opanasenko, N.V., 2001. Absolute calibration of the Clementine UVVIS data: comparison with ground-based observation of the Moon. Solar System Res. 35 (1), 29–34.

Shkuratov, Yu.G., Stankevich, D., Kaydash, V., Omelchenko, V., Pieters, C., Pinet, C., Chevrel, S., Daydou, Y., Foing, B., Sodnik, Z., Josset, J.-L., Taylor, L., Shevchenko, V., 2003a. Composition of the lunar surface as will be seen from SMART-1: a simulation using Clementine data. J. Geophys. Res. 108 (E4).

Shkuratov, Yu.G., Pieters, C.M., Omelchenko, V.G., Stankevich, D.G., Kaydash, V., Taylor, L., 2003b. Estimates of the lunar surface composition with Clementine images and LSCC data. Lunar Planet. Sci., (34th Abstract #1258. LPI Houston).

Shkuratov, Yu.G., Kaydash, V.G., Pieters, C.M., 2005a. Lunar clinopyroxene and plagioclase: surface distribution and composition. Solar System Res. 39, 255–266.

Shkuratov, Yu.G., Kaydash, V., Stankevich, D., Pinet, P., Chevrel, S., Daydou, Y., 2005b. Derivation of elemental abundance maps at intermediate resolution from optical interpolation of Lunar Prospector Gamma-Ray Spectrometer data. Planet. Space Sci. 53, 1287–1301.

Shkuratov, Yu.G., Kaydash, V.G., Starukhina, L.V., Pieters, C., 2007. Lunar surface agglutinates: mapping composition anomalies. Solar System Res. 41, 177–185.

Staid, M.I., Pieters, C.M., 2001. Mineralogy of the last lunar basalts: Results from Clementine. J. Geophys. Res. 106 (E11), 27887–27900.

Taylor, L.A., Pieters, C.M., Morris, R.V., Keller, L.P., McKay, D.S., 2001. Lunar mare soils: space weathering and the major effects of surface-correlated nano-phase Fe. J. Geophys. Res. 106 (E11), 27,985–28,000.

Taylor, L.A., Pieters, C.M., Patchen, A., Taylor, D.H., Morris, R.V., Keller, L.P., McKay, D.S., 2003. Mineralogical characterization of lunar highland soils. In: Lunar and Planetary Science Conference XXXIV, CD-ROM, Abstract #1774. LPI Houston.